

Research Interests

- Statistical Learning under Complex Data Conditions**
Semi-supervised learning, informative missingness, mixture models, robust estimation, kernel and regularisation methods
- Computational Statistics and Optimization for Learning**
Efficient optimisation for statistical models, high-dimensional and non-convex problems, stochastic/iterative optimisation, evolutionary and swarm-based methods
- Time-series, Forecasting, and Causal Inference**
Statistical modelling and uncertainty quantification in energy, environmental, and socio-economic systems; probabilistic forecasting; causal mechanisms in dynamical systems

Education

- 2022 **Doctor of Philosophy in Statistics**, *Queensland University of Technology*, Brisbane, Australia
 2017 **Master of Applied Statistics**, *Lanzhou University*, Lanzhou, PR China
 2014 **Bachelor of Economics in Economic Statistics**, *Anhui University*, Hefei, PR China

PhD Thesis

- Title *Statistical Support Vector Machines with Optimizations*
 (The 2022 Executive Dean's Commendation for Outstanding Doctoral Thesis)
 Supervisors Professors You-Gan Wang, Kevin Burrage, & Yu-Chu Tian

Employments

- Feb 2025– **Research Fellow**, *School of Mathematics and Physics, The University of Queensland*, St Lucia, Australia, with Professor Geoffrey J. McLachlan (Fellow of the Australian Academy of Science)
 Dec 2022– Jan 2025 **Research Fellow**, *Faculty of Education and Arts, Australian Catholic University*, North Sydney, Australia, with Professor Herbert W. Marsh (Fellow of the Academy of the Social Sciences in Australia and the British Academy of Social Sciences)
 Jul 2022– Nov 2022 **Associate Lecturer**, *School of Mathematical Sciences, Queensland University of Technology*, Brisbane, Australia, (short contract)
 Jul 2021– Nov 2021 **Sessional Academic**, *School of Mathematical Sciences, Queensland University of Technology*, Brisbane, Australia
 Jun 2018– Dec 2018 **Research Assistant**, *UQ Business School, The University of Queensland*, St. Lucia, Australia, with Dr Min Zhu

Refereed Articles (*Correspondence; Led study design & writing)

- Jinran Wu**, You-Gan Wang, and Geoffrey J McLachlan. Informative missingness and its implications in semi-supervised learning. *The Innovation Informatics*, 2:100033, 2026.
- Jinran Wu**, Xin Tian, You-Gan Wang, Tong Li, Qingyang Liu, Yayong Li, Lizhen Cui, Zhuangcai Tian, Jing Xu, Xianzhou Lyu, et al. AI ethics in geoscience: Toward trustworthy and responsible innovation. *Geography and Sustainability*, 7(1):100414, 2026.
- Jinran Wu**, Xin Tian, Qingyang Liu, Tong Li, Chanjuan Liu, Jing Xu, and Huida Zhao. Generative artificial intelligence and urban transport resilience: Rethinking sustainable mobility. *Safety Science*, 198:107148, 2026.

- [4] **Jinran Wu**, Herbert W. Marsh, Jiesi Guo, Johnmarshall Reeve, Reinhard Pekrun, Theresa Dicke, Hye-Ren Jang, and Geetanjali Basarkod. Mapping the intellectual landscape of educational psychology: Citation rankings and network structures of 60 journals, scholars, and institutions. *Learning and Individual Differences*, 125:102816, 2026.
- [5] Yongjun Xu, Zezhi Shao, Xin Liu, ..., **Jinran Wu**, et al. AI for science: Progress, challenges, and perspectives. *The Innovation*, 7, 2026.
- [6] Huanchao Zhou, Fang Wang, You-Gan Wang, **Jinran Wu**, Jiang Hu, Geoffrey J McLachlan, Zhi-dong Bai, and Jianghao Li. Signal plus noise type matrix in random matrix theory: A review. *Wiley Interdisciplinary Reviews: Computational Statistics*, 18(3):e70071, 2026.
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- [8] Yujia Huang, **Jinran Wu**, Zhe Ding, and Xi'an Li. Normalized Fourier induced coupled PINNs to solve the Dirichlet biharmonic equations in a large-scale domain. *Frontiers of Computer Science*, 20(7):2007354, 2026.
- [9] Xi'an Li, **Jinran Wu**, Xiao Ning, Yangshuai Wang, and Lei Zhang. Bayesian physics-informed neural networks to solve the partial differential equations with noise and incomplete constraints. *The Innovation Informatics*, 2:100039, 2026.
- [10] Tian Chen, Zhuangcai Tian, Xi'an Li, Jingxin Wu, Xiujun Guo, Yumeng Zhang, Haorui Wang, and **Jinran Wu**. Physics-informed neural networks help to strongly reduce uncertainty in ocean state forecasting. *The Innovation Informatics*, 2:100051, 2026.
- [11] Jing Xu, Siyuan Liu, **Jinran Wu**, Yuming Mo, Kun Wang, Qingfeng Ji, You-Gan Wang, and Ling Li. Multi-head attention driven aggregation-differentiation network for probabilistic groundwater depth forecasting and depth-stratified early warning: A multi-site efficient framework. *Environmental Modelling & Software*, 199:106934, 2026.
- [12] Chaojun Zou, Fang Wang, **Jinran Wu**, You-Gan Wang, and Xinghui Zhu. Rapid rapeseed identification with dual-graph discriminant analysis in hyperspectral imaging. *Chemometrics and Intelligent Laboratory Systems*, 277:105813, 2026.
- [13] Jichao Ma, Dandan Liu, Xi'an Li, Yayong Li, and **Jinran Wu***. A spatial-temporal normalization framework for fourier-enhanced physics-informed deep learning in large-scale wave propagation. *Discover Computing*, 2026.
- [14] Shuaiying Zhang, Zhen Dong, Huadong Lin, Zhendong Zhang, **Jinran Wu**, Sinong Quan, Wentao An, Tong Li, and Rajiv Pandey. Integrating linear and circular polarization features for polsar land cover classification with deep learning. *International Journal of Applied Earth Observation and Geoinformation*, 146:105090, 2026.
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- [17] Xin Tian, Qingyang Liu, Ram C. Dalal, Tong Li, **Jinran Wu***, Geoffrey J. McLachlan, Scott Chapman, and Yash P. Dang. Versioned soil organic carbon maps for climate-resilient and sustainable cities: A ridge-penalised linear mixed-effects ensemble of MIR-based predictions. *Physics and Chemistry of the Earth*, 143:104390, 2026.
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- [39] Zhesen Cui, Qingbin Zhang, Zhixing Zhao, Maosheng Yang, Pan Kong, Shuobao Cai, Jinran Zhang, Jing Xu, **Jinran Wu***, Rajib Pandey, and Tong Li. A two-stage probabilistic forecasting framework for dam displacement: Evidence from a Chinese watershed. *Journal of Hydrology: Regional Studies*, 62:102822, 2025.
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Review Services

Grants

- DAAD Postdoc-Net-AI under German Academic Exchange Service

Journals

- | | |
|---|---|
| ○ Agronomy | ○ Ain Shams Engineering Journal |
| ○ Algorithms | ○ Applied Artificial Intelligence |
| ○ Applied Intelligence | ○ Applied Mathematics and Mechanics |
| ○ Applied Sciences | ○ Artificial Intelligence Review |
| ○ Asia Pacific Journal of Marketing and Logistics | ○ Automatika |
| ○ Biology and Fertility of Soils | ○ CAAI Transactions on Intelligence Technology |
| ○ Canadian Journal of Chemical Engineering | ○ Carbon Footprints |
| ○ Clinical Cardiology | ○ Communications in Computational Physics |
| ○ Complex & Intelligent Systems | ○ Concurrency and Computation: Practice and Experience |
| ○ Control Engineering Practice | ○ Cybernetics and Systems |
| ○ Data Science and Management | ○ Earth's Future |
| ○ Earthquake Engineering and Resilience | ○ Electric Power Systems Research |
| ○ Electronics | ○ Energies |
| ○ Energy Efficiency | ○ Energy Efficiency First |
| ○ Energy Exploration & Exploitation | ○ Engineered Science |
| ○ Engineering Applications of Artificial Intelligence | ○ Engineering Applications of Computational Fluid Mechanics |
| ○ Engineering Computations | ○ Engineering Optimization |

- Engineering Research Express
- Environmental Engineering Science
- Environmental Modeling and Assessment
- Environmental Science: Advances
- Expert Systems with Applications
- Forecasting
- Frontiers of Computer Science
- Frontiers of Structural and Civil Engineering
- Human and Ecological Risk Assessment
- IEEE Antennas and Wireless Propagation Letters
- IEEE Geoscience and Remote Sensing Letters
- IEEE Internet of Things Journal
- IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing
- IEEE Transactions on Artificial Intelligence
- IEEE Transactions on Cybernetics
- IEEE Transactions on Emerging Topics in Computational Intelligence
- IEEE Transactions on Geoscience and Remote Sensing
- IEEE Transactions on Intelligent Transportation Systems
- IEEE Transactions on Knowledge and Data Engineering
- IEEE Transactions on Network and Service Management
- IEEE Transactions on Pattern Analysis and Machine Intelligence
- IEEE Transactions on Smart Grid
- IEEE Transactions on Vehicular Technology
- IEEE/ACM Transactions on Audio, Speech, and Language Processing
- IET Communications
- IET Generation, Transmission & Distribution
- IET Renewable Power Generation
- Information Processing in Agriculture
- Integrated Environmental Assessment and Management
- International Journal for Numerical Methods in Engineering
- International Journal of Ambient Energy
- International Journal of Computer Mathematics
- International Journal of Forecasting
- International Journal of Information Technology & Decision Making
- Entropy
- Environmental Impact Assessment Review
- Environmental Science and Pollution Research
- Europhysics Letters
- F1000Research
- Frontiers in Energy Research
- Frontiers of Information Technology & Electronic Engineering
- Geo-Spatial Information Science
- IEEE Access
- IEEE Computational Intelligence Magazine
- IEEE IAS Publications
- IEEE Journal of Biomedical and Health Informatics
- IEEE Sensors Journal
- IEEE Transactions on Automation Science and Engineering
- IEEE Transactions on Dependable and Secure Computing
- IEEE Transactions on Fuzzy Systems
- IEEE Transactions on Industrial Informatics
- IEEE Transactions on Intelligent Vehicles
- IEEE Transactions on Microwave Theory and Techniques
- IEEE Transactions on Neural Networks and Learning Systems
- IEEE Transactions on Reliability
- IEEE Transactions on Systems, Man, and Cybernetics: Systems
- IEEE Wireless Communications Letters
- IEEE/CAA Journal of Automatica Sinica
- IET Electric Power Applications
- IET Power Electronics
- IET Smart Grid
- Innovation Informatics
- International Journal for Numerical and Analytical Methods in Geomechanics
- International Journal of Adaptive Control and Signal Processing
- International Journal of Computational Intelligence Systems
- International Journal of Environmental Science and Technology
- International Journal of Green Energy
- International Journal of Modelling and Simulation

- International Journal of Numerical Methods for Heat & Fluid Flow
- International Journal of Urban Sciences
- JMIR Public Health and Surveillance
- Journal of Applied Statistics
- Journal of Chemical & Engineering Data
- Journal of Computer Science and Technology
- Journal of Ecohydraulics
- Journal of Field Robotics
- Journal of Groundwater Science and Engineering
- Journal of Marine Engineering and Technology
- Journal of Modern Power Systems and Clean Energy
- Journal of Vibration and Control
- Machine Learning: Science and Technology
- Marine Georesources and Geotechnology
- Measurement Science and Technology
- Medicine
- Network: Computation in Neural Systems
- Neurocomputing
- Nonlinear Dynamics
- Ocean Engineering
- Personnel Review
- Physica Scripta
- Physics of Fluids
- Results in Chemistry
- Science Progress
- Ships and Offshore Structures
- SN Computer Science
- Spectroscopy Letters
- Statistics in Biosciences
- Symmetry
- Technological Forecasting and Social Change
- Transactions on Emerging Telecommunications Technologies
- International Journal of Systems Science
- iScience
- Journal of Ambient Intelligence and Humanized Computing
- Journal of Bionics Engineering
- Journal of Computational Social Science
- Journal of Earth Science
- Journal of Environmental Management
- Journal of Geophysical Research: Oceans
- Journal of Hospitality Marketing & Management
- Journal of Mechanical Engineering Science
- Journal of Supercomputing
- Machine Learning
- Malaysian Journal of Mathematical Sciences
- Mathematics
- Mechanics of Advanced Materials and Structures
- Microchemical Journal
- Neural Computing & Applications
- New England Journal of Medicine
- Numerical Methods for Partial Differential Equations
- PeerJ Computer Science
- Petroleum Science and Technology
- Physical Chemistry Chemical Physics
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- Safety Science
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Conference Papers and Presentations

- [12] Qiu, X., Yuan, J., **Wu, J.**, Li, W., Guo, J., & Leung, F. (2026). Leveraging Machine Learning for Clustered Data: Model Development and Application to Examining the Impact of COVID-19 on Student Mathematics Achievement. *International Meeting of the Psychometric Society 2026*. Seoul, Republic of Korea
- [11] Zhang, H., Chen, Y., Liu, X., **Wu, J.**, Liao, Y., & Li, Y. (2025). A margin enhanced data augmentation method for imbalanced credit default prediction. *2025 IEEE International Conference on Big Data (IEEE BigData 2025)*. Macau, China
- [10] Li, X., **Wu, J.**, Liu, Y., Xu, J. & Tai, X. (2025). Fourier induced DNN to assist the channel estimation and signal detection in OFDM systems. *18th UK-Europe-China Workshop on Millimetre Waves and Terahertz Technologies (UCMMT2025)*. Nanjing, China

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- [6] Cui, Z., **Wu, J.**, & Lian, W. (2021). A modified gated recurrent unit neural network for crude oil price forecasting. *The 2021 International Academic Conference on Machine Learning, Big Data, and Statistics, Financial Information Technology (ICMBSF 2021)*. Brisbane, Australia
- [5] Gao, Y., Qian, C., Tao, Z., Zhou, H., **Wu, J.**, & Yang, Y. (2020). Improved whale optimisation algorithm via cellular automata. *The 2020 International Conference on Progress in Informatics and Computing (PIC-2020)*. Shanghai, China
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- [2] **Wu, J.**, & Ding, Z. (2020). Improved grey model by dragonfly algorithm for Chinese tourism-demand forecasting. *The 33th International Conference on Industrial, Engineering & Other Applications of Applied Intelligent Systems*. Kitakyushu, Japan
- [1] **Wu, J.**, & Wang, Y.-G. (2019). Time series regression modelling using phase space reconstruction and deep neural networks. *The 20th INFORMS Applied Probability Society Conference*. Brisbane, Australia

Professional Memberships and Academic Services

Jul 2024–	Academic Editor of the <i>PloS One</i>
May 2026–	Associate Editor of <i>Journal of Electrical Systems and Information Technology</i>
Jan 2026–	Editorial Board Member of the <i>Applied Mathematics and Statistics</i>
Oct 2025–	Youth Editor of <i>The Innovation Informatics</i>
Dec 2023–	Youth Editor of <i>Deep Underground Science & Engineering</i>
Jun 2026	Guest Editor, the Special Issue on “Particulate and Granular Flow in the Ocean” of <i>Physics of Fluids</i>
Jan 2026	Guest Editor, the Special Issue on “Artificial Intelligence for Biochar Applications in Agricultural, Forest, and Grassland Ecosystems: From Soil Carbon to Ecosystem Sustainability” of <i>Biochar</i>
Sep 2025	Guest Editor, the Special Issue on “Artificial Intelligence in Physical Geography: Theory, Methods, and Frontiers of Practice” of <i>Geographical Science</i> [In Chinese]
Jun 2025	Guest Editor, the Special Issue on “Foundation Models and Artificial Intelligence in Geography and Sustainability” of <i>Geography and Sustainability</i>
May 2025	Guest Editor, the Special Issue on “Impact of Generative Artificial Intelligence on Transportation Safety and Resilience” of <i>Safety Science</i>
Apr 2025	Guest Editor, the Special Issue on “Stability and Persistence of Soil Organic Carbon Pool in a Changing World” of <i>Biology and Fertility of Soils</i>
Mar 2025	Member of the Program Committee of 26th International Conference on Artificial Intelligence in Education (AIED 2025), Palermo, Italy
Nov 2024	Member of the Program Committee of the Australasian Data Science and Machine Learning Conference (AusDM'24), Melbourne, Australia
Jul 2024	Member of the Program Committee of the 17th International Conference on Educational Data Mining (EDM 2024), Atlanta, USA

Sep 2024	Guest Editor, the Special Issue on “Deep-Sea Sediment Transport and Mining” of <i>Frontiers in Earth Science</i>
Jan 2024	Guest Editor, the Special Issue on “Data-Driven Approaches for Efficient Smart Grid Systems” of <i>Frontiers in Energy Research</i>
Sep 2023	Session Convener, the XIV Congress of the International Association for Engineering Geology and the Environment, Chengdu, China
Jan 2023	Guest Editor, the Special Issue on “Machine Learning in Environmental Modelling” of <i>Environmental Modeling and Assessment</i>
Jul 2022–	Review Editor, Editorial Board of Machine Learning and Artificial Intelligence (speciality section of <i>Frontiers in Big Data</i> and <i>Frontiers in Artificial Intelligence</i>)

Teaching Experience

2026se1	Instructor for MATH7014 (Minor Project), UQ
2025se2	Instructor for MATH7012 (Project or Thesis), UQ
2025se2	Guest Lecturer for STAT4401 (Advanced Statistics), UQ
2024se2	Coordinator for EDCP600 (Interpreting Literature and Data), ACU
2022se2	Coordinator for MXB343 (Modelling Dependent Data), QUT
2022se1	Guest Lecturer for MXN441 (Advanced Statistical Inference and Modelling), QUT
2021se2	Tutor for MXB343 (Modelling Dependent Data), QUT
2021se1	Guest Lecturer for MXN441 (Advanced Statistical Inference and Modelling), QUT

Grants and Awards

2024	The 2024 Early Career Researcher Grant, Faculty of Education & Arts, ACU Be awarded the amount of AUD\$5,940.24
2023	The 2022 Executive Dean’s Commendation for Outstanding Doctoral Thesis, Faculty of Science, QUT
2022	The Chinese Government Award for Outstanding Self-financed Students Abroad The highest governmental award to Chinese graduate students studying abroad
2020	The QUT Science and Engineering Faculty’s High Achiever Award
2018	The Australian Government Research Training Program Scholarship (International)
2018	The ACEMS Top Up Scholarship